

$$\textcircled{2} \quad B = \mu_0 n I$$

$$\phi = \int B dA$$

$$= \int \mu_0 n I dA$$

$$= \mu_0 n I \cdot A$$

$$= \mu_0 n I (\pi r^2)$$

$$r = 0.02 \text{ m}$$

$$\mathcal{E} = -N \frac{d\phi}{dt}$$

$$= -15 \left[\mu_0 n \pi r^2 \frac{dI}{dt} \right]$$

$$= -15 \cdot 4\pi \cdot 10^{-7} \cdot 1000 \cdot \pi \cdot (0.02)^2 \cdot 5 \cdot \cos(120t) \cdot 120$$

$$= -0.014 \cos(120t) \text{ [V]}$$

$$= -14 \cos(120t) \text{ [mV]}$$